Finite dimensional terminal gradient bounds and sharp performance under relative errors

Abstract

We study constant-step gradient descent on smooth convex functions with arbitrary deterministic relative gradient errors below one. In every fixed finite dimension, the terminal squared gradient admits a class-uniform $O(1/N)$ bound, normalized by the initial function gap, precisely when the normalized step lies in $0 < h < 2/(1+\delta)$. The proof bounds gradient amplification by selecting the last vertices below geometric thresholds, separating their gradient directions through convex support inequalities, and applying spherical packing. A staged family in logarithmically growing dimensions rules out a dimension-independent $O(1/N)$ bound for $0 < h \leq 1$. The resulting exponential lower and generally double-exponential upper dimension costs leave a substantial gap. A planar ray argument gives a dimension-specific estimate at smaller steps and a continuous-time counterpart. For the minimum gradient over the updated iterates, a local interpolation certificate with a telescoping correction and explicit attaining trajectories determine the exact asymptotic coefficient and its optimal constant step. The one-step curve is included with its known intermediate extremizer and the associated optimal-step geometry.

Keywords: inexact gradient descent; relative error; terminal gradient; finite dimensional geometry; smooth convex interpolation

2020 Mathematics Subject Classification. Primary 65K10; Secondary 90C25, 90C60.

1 The performance problem

Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and convex, with an L-Lipschitz gradient, where $L > 0$ and d is any positive integer. Assume that f attains its minimum f_*. Fix $0 < \delta < 1$ and a normalized step $h \geq 0$. We consider

$$x_{k+1} = x_k - \frac{h}{L} d_k, \qquad \|d_k - \nabla f(x_k)\| \leq \delta \|\nabla f(x_k)\|. \tag{1}$$

The directions are arbitrary deterministic vectors satisfying the displayed condition at each update. The same h is used throughout a run.

For $N \geq 1$, define

$$\mathcal{W}_N(h, \delta) = \sup \frac{\min_{1 \leq k \leq N} \|\nabla f(x_k)\|^2}{L(f(x_0) - f_*)}, \tag{2}$$

where the supremum is over all the preceding data with $f(x_0) > f_*$. Thus the dimension, objective, initial point and errors may depend on N. The initial point is excluded from the minimum, and the final iterate is included. For $N = 1$, this is the terminal-gradient criterion.

For the terminal gradient in a fixed dimension, define

$$\mathcal{W}_{N,d}^{\mathrm{last}}(h,\delta)=\sup\frac{\|\nabla f(x_N)\|^2}{L(f(x_0)-f_*)},\qquad M_d(h,\delta)=\sup_{N\ge 1}N\mathcal{W}_{N,d}^{\mathrm{last}}(h,\delta). \tag{3}$$

Here the first supremum is restricted to $\mathbb{R}^d$, with positive initial gap. The objective, initial point and error history may still depend on N; only d, h, δ are fixed. Theorem 1 proves that M_d is finite for every finite d exactly on the open stability interval $0 < h < 2/(1+\delta)$. It applies to every initial point and every admissible error history, without a bounded-trajectory or strong-convexity assumption. In particular, it includes the normalized unit step in the plane and in every higher fixed dimension.

The geometric proof uses a uniform descent-to-path-length estimate. On an interval whose gradients stay above their initial magnitude, convex supporting inequalities force sufficiently different gradient magnitudes to have separated directions. Last exits from increasing thresholds produce such intervals without assuming monotonic gradient magnitudes. A finite-dimensional sphere can contain only finitely many separated directions, which bounds all subsequent gradient amplification. Summed descent then gives the terminal rate. Section 6 constructs a Huber family with dimension $2+\log_2 N$ showing that the constant cannot be uniform in dimension when $0 < h \le 1$. Its direct consequence is an exponential lower bound for M_d; the packing argument gives a generally double-exponential upper bound. Their dimension dependence is not sharp.

For this error model, Vernimmen and Glineur [1] derive a three-regime one-step upper bound and supply scalar examples in the two outer regimes. Their Conjecture 2.6 asks for bivariate attainment in the intermediate regime and remains in the official accepted manuscript. Zhao [2, Theorem 2.1] supplies a public analytic resolution throughout that interval, together with a Huber–quadratic extremizer; his Section 6 also determines the one-dimensional envelope. Intermediate attainment is therefore treated as established background here. Section 3 gives a self-contained derivation in the present parameterization, together with complete certificates and the associated one-step optimization geometry. The theorem and conjecture numbering refers to the public Vernimmen–Glineur version cited in the bibliography.

The multiple-step result of [1] concerns the minimum historical gradient. Sections 4 and 5 determine the exact asymptotic coefficient for the specific criterion (2), its minimax constant step and convergence of asymptotically minimizing step sequences. Those statements do not determine the exact finite-N curve or the sharp fixed-dimensional terminal constant. The initial condition throughout is a function gap, rather than a distance to a minimizer.

For the last gradient, Vernimmen's thesis [7, Section 2.2] already studies one-step amplification, gives a non-tight terminal estimate and compares last and best gradients numerically. We retain that ancestry. Angle and scaling conditions also underlie the strongly convex function-value contraction results of Uschmajew and Vandereycken [3].

Gradient oscillations for strongly convex quadratics with exact line search are analyzed in [4]. Those performance criteria and assumptions differ from the class-uniform terminal estimate in (3).

Planar geometry has an established role in descent: Daniilidis, Deville and Durand-Cartagena [8, Theorem 6.4] prove rectifiability of bounded planar curves satisfying a cone condition. Exact-gradient path-length bounds in [9] depend on spatial distances. Section 7 gives the dimension-specific planar argument based on newly visited gradient rays, including its continuous-time version. The higher-dimensional proof does not extend that ray-order argument formally: it uses convex supports and separation across last exits. Smooth convex interpolation [5] supplies the algebraic certificates in the performance sections. Earlier robustness results for strongly convex objectives, including [6], address a different problem.

2 Terminal gradients in every fixed dimension

We first establish the fixed-dimensional result, independently of the exact one-step curve. Write $g_k = \nabla f(x_k)$ and $e_k = d_k - g_k$.

Theorem 1 (Fixed-dimensional stability classification)**.** *For every finite $d \geq 1$, every $0 < \delta < 1$ and every $h \geq 0$,*

$$M_d(h,\delta) < \infty \quad \Longleftrightarrow \quad 0 < h < \frac{2}{1+\delta}. \tag{4}$$

For h in this open interval, define

$$\begin{aligned} c &= \sqrt{1-\delta^2}\left(1 - \frac{h(1+\delta)}{2}\right), \qquad A = 1 + h(1+\delta), \qquad q = 2A, \\ K_d &= \left\lceil (1+4/c)^d \right\rceil, \qquad B_d = q^{K_d}, \end{aligned} \tag{5}$$

and

$$\alpha_h = h - \frac{h^2(1+\delta^2)}{2} - h|1-h|\delta = \begin{cases} h(1-\delta)\left(1 - \dfrac{h(1-\delta)}{2}\right), & h \leq 1, \\ h(1+\delta)\left(1 - \dfrac{h(1+\delta)}{2}\right), & h \geq 1. \end{cases} \tag{6}$$

Then $c, \alpha_h > 0$. Every admissible run satisfies

$$\|g_j\| \leq B_d \|g_i\| \qquad (0 \leq i \leq j). \tag{7}$$

For every $N \geq 1$,

$$\mathcal{W}^{\mathrm{last}}_{N,d}(h,\delta) \leq \frac{1}{\frac{1}{2} + \alpha_h N / B_d^2}, \qquad M_d(h,\delta) \leq \frac{B_d^2}{\alpha_h}. \tag{8}$$

For $d = 3$, the integer K_d may be replaced by $\lceil 64/c^2 \rceil$.

The constants are deliberately coarse. On $0 < h \le 1/(1+\delta)$, one has $q \le 4$ and $c \ge \sqrt{1-\delta^2}/2$, so a simple valid three-dimensional amplification constant is

$$B_3 = 4^{\lceil 256/(1-\delta^2)\rceil}. \tag{9}$$

Theorem 1 rules out any finite failure dimension. Section 6 shows why its dependence on d cannot be discarded.

2.1 Convex supports separate gradient directions

The following lemma applies to gradients of one convex function. The lower bound on the intermediate gradient magnitudes converts objective decrease into a path-length budget.

Lemma 1 (Separation on a protected interval)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and convex, and let $g_k = \nabla f(x_k)$. Suppose that a finite point sequence satisfies*

$$f(x_k) - f(x_{k+1}) \ge c\|g_k\|\|x_{k+1} - x_k\|, \qquad c > 0. \tag{10}$$

If $i < j$, $r_i = \|g_i\| > 0$, and

$$\|g_k\| \ge r_i \quad (i \le k < j), \qquad a = \frac{\|g_j\|}{r_i} > 1,$$

then

$$\left\| \frac{g_i}{\|g_i\|} - \frac{g_j}{\|g_j\|} \right\| \ge c\left(1 - \frac{1}{a}\right). \tag{11}$$

Proof. Write $\Delta = f(x_i) - f(x_j)$ and $v = x_i - x_j$. Summing (10) gives

$$\|v\| \le \sum_{k=i}^{j-1} \|x_{k+1} - x_k\| \le \frac{\Delta}{c r_i}. \tag{12}$$

The supporting inequalities at the two endpoints give

$$\langle g_i, v\rangle \ge \Delta \ge \langle g_j, v\rangle. \tag{13}$$

Consequently,

$$\begin{aligned}(a-1)\Delta &\le \langle a g_i - g_j, v\rangle \\ &\le a r_i \left\| \frac{g_i}{r_i} - \frac{g_j}{a r_i} \right\| \|v\| \\ &\le \frac{a}{c} \left\| \frac{g_i}{r_i} - \frac{g_j}{a r_i} \right\| \Delta.\end{aligned}$$

Here $\Delta > 0$: if $\Delta = 0$, (12) makes every intervening displacement zero, so $g_i = g_j$, contrary to $a > 1$. Division proves (11). ∎

2.2 Last exits and spherical packing

A first visit to a gradient threshold need not leave later gradients above that threshold. The last vertex at or below the threshold supplies precisely the lower bound required in Lemma 1.

Lemma 2 (Uniform amplification from last exits)**.** *In addition to* (10)*, assume*

$$\|g_{k+1}\| \le A\|g_k\|, \qquad A \ge 1. \tag{14}$$

Let K be a nonnegative integer such that every subset of $\mathbb{S}^{d-1}$ with pairwise Euclidean distances strictly greater than $c/2$ has at most K points. Then

$$\|g_j\| \le (2A)^K \|g_i\| \qquad (0 \le i \le j).$$

Proof. Restart at i and relabel the finite interval as $0, \dots, N$. If $g_0 = 0$, (14) makes every subsequent gradient zero. Otherwise put $r_0 = \|g_0\| > 0$ and $q = 2A$. Suppose that $\|g_N\| > q^K r_0$. For $\ell = 0, \dots, K$, define

$$s_\ell = \max\{k \in \{0, \dots, N-1\} : \|g_k\| \le q^\ell r_0\}. \tag{15}$$

Each set contains 0, and its threshold is below the terminal magnitude. The last-exit property and (14) imply

$$\frac{q^\ell r_0}{A} < \|g_{s_\ell}\| \le q^\ell r_0, \qquad \|g_k\| > q^\ell r_0 \quad (s_\ell < k \le N). \tag{16}$$

Indeed, the gradient at $s_\ell + 1$ exceeds the threshold and is at most $A\|g_{s_\ell}\|$.

For $\ell < m$, the threshold sets are nested, so $s_\ell \le s_m$. Moreover,

$$\frac{\|g_{s_m}\|}{\|g_{s_\ell}\|} > \frac{q^{m-\ell}}{A} \ge 2,$$

which excludes equal indices. Every intermediate gradient in $[s_\ell, s_m)$ has magnitude at least $\|g_{s_\ell}\|$, by (16). Lemma 1 therefore shows that the $K+1$ unit vectors

$$\frac{g_{s_0}}{\|g_{s_0}\|}, \dots, \frac{g_{s_K}}{\|g_{s_K}\|}$$

have pairwise distances strictly greater than $c/2$. This contradicts the definition of K. ∎

For unit vectors separated by more than $\varepsilon > 0$, the open Euclidean balls of radius $\varepsilon/2$ around them are disjoint and lie in the ball of radius $1 + \varepsilon/2$. Comparing volumes bounds their number by $(1 + 2/\varepsilon)^d$. Taking $\varepsilon = c/2$ gives the integer K_d in (5). In three dimensions, disjoint spherical caps of angular radius $\arcsin(\varepsilon/2)$ give

$$\#\{\text{points}\} \le \frac{2}{1 - \sqrt{1 - \varepsilon^2/4}} \le \frac{16}{\varepsilon^2} = \frac{64}{c^2}, \tag{17}$$

where the last equality uses $\varepsilon = c/2$. Here $0 < c < 1$, so the cap radii are well defined. Taking the ceiling provides a valid integer packing bound.

2.3 Descent efficiency and the terminal estimate

Proof of Theorem 1. For $d = g + e$ with $\|e\| \le \delta\|g\|$, the relative-error ball implies

$$\|d\|^2 \le (1+\delta)\langle g, d\rangle, \qquad \langle g, d\rangle \ge \sqrt{1-\delta^2}\|g\|\|d\|. \tag{18}$$

The first inequality follows from

$$\|d\|^2 - (1+\delta)\langle g, d\rangle = -\delta\|g\|^2 + (1-\delta)\langle g, e\rangle + \|e\|^2 \le 0.$$

For the second, if $g \neq 0$, then $\langle g, d\rangle \ge (1-\delta)\|g\|^2 > 0$. The component of g perpendicular to d is the negative of the corresponding component of e, and hence has norm at most $\delta\|g\|$. The positive component along d is therefore at least $\sqrt{1-\delta^2}\|g\|$. If $g = 0$, the error bound gives $d = 0$ and both inequalities hold.

Smoothness and (18) yield

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\ge \frac{h}{L}\left(\langle g_k, d_k\rangle - \frac{h}{2}\|d_k\|^2\right) \\ &\ge \frac{h}{L}\left(1 - \frac{h(1+\delta)}{2}\right)\langle g_k, d_k\rangle \\ &\ge c\|g_k\|\|x_{k+1} - x_k\|. \end{aligned}$$

This proves (10) on the entire stable interval. It requires descent only at the vertices, not on every point of each interpolating segment. Lipschitz continuity also gives

$$\|g_{k+1}\| \le \|g_k\| + L\|x_{k+1} - x_k\| \le [1 + h(1+\delta)]\|g_k\|,$$

so (14) holds with A from (5). Lemma 2 and the packing bounds prove (7).

Expanding the smooth descent inequality once more gives

$$\begin{aligned} L[f(x_k) - f(x_{k+1})] &\ge \left(h - \frac{h^2}{2}\right)\|g_k\|^2 + h(1-h)\langle g_k, e_k\rangle - \frac{h^2}{2}\|e_k\|^2 \\ &\ge \alpha_h\|g_k\|^2. \end{aligned} \tag{19}$$

The two expressions in (6) show that $\alpha_h > 0$. For every $k < N$, restarting (7) at k gives $\|g_k\|^2 \ge \|g_N\|^2/B_d^2$. Summing (19) and using the smooth convex bound $L[f(x_N) - f_*] \ge \|g_N\|^2/2$ yields

$$L[f(x_0) - f_*] \ge \left(\frac{1}{2} + \frac{\alpha_h N}{B_d^2}\right)\|g_N\|^2.$$

This proves (8) for each objective, initialization, history and horizon before either supremum is taken. A zero-gradient vertex is stationary by the error condition, so this also covers zero initial gaps and zero gradients wherever the unnormalized estimate is used.

For necessity, a one-dimensional quadratic embedded in $\mathbb{R}^d$ suffices. Let $f(x) = Lx_1^2/2$, $x_0 = (1, 0, \ldots, 0)$ and $e_k = \delta g_k$. Then

$$x_{k+1,1} = [1 - h(1+\delta)]x_{k,1}, \qquad \frac{\|g_N\|^2}{L(f(x_0) - f_*)} = 2|1 - h(1+\delta)|^{2N}.$$

For $h = 0$ or $h \ge 2/(1+\delta)$, this ratio is at least two, so $M_d = \infty$. At the positive stability endpoint the scalar iterate reflects with constant gradient magnitude. This proves (4). ∎

3 The exact one-step curve

Put $D = \delta^2$ and define

$$q_R = \frac{2 + \delta - \sqrt{4 - 3\delta^2}}{2}, \quad h_L = \frac{3}{2(1+\delta)}, \quad h_R = \frac{3\delta + 2 - \sqrt{4 - 3\delta^2}}{2\delta(1+\delta)}. \tag{20}$$

For $q_R \le q \le \delta$, write

$$V = 2q - q^2 - D, \quad U = 3q - q^2 - 2D, \quad h(q) = \frac{(1-q)U}{(1-D)V}, \quad \rho(q) = \frac{(1-q)U}{2q(1-D)}. \tag{21}$$

Theorem 2. *The map $q \mapsto h(q)$ is a continuous strictly decreasing bijection from $[q_R, \delta]$ onto $[h_L, h_R]$. For $0 \le h \le 2/(1+\delta)$,*

$$\mathcal{W}_1(h, \delta) = \begin{cases} [h(1-\delta) + \frac{1}{2}]^{-1}, & 0 \le h \le h_L, \\ [\rho(q) + \frac{1}{2}]^{-1}, \quad h = h(q), & h_L \le h \le h_R, \\ 2[1 - h(1+\delta)]^2, & h_R \le h \le 2/(1+\delta). \end{cases} \tag{22}$$

The endpoint values agree. Every value is attained by a globally smooth convex function in dimension at most two.

The outer regimes and the intermediate upper bound are those of [1, Theorems 2.3–2.4]. Intermediate attainment, and hence resolution of its Conjecture 2.6, is also established by Zhao [2]. We retain the following complete derivation in the parameter q to connect the exact curve with its optimal step; the upper certificate and global attaining function make the equality directly verifiable.

3.1 Interpolation certificates

Dividing f and the directions by L, and subtracting f_*, reduces the analysis to $L = 1$ and $f_* = 0$. At one step write $g = \nabla f(x)$, $g_+ = \nabla f(x_+)$ and $e = d - g$, where $x_+ = x - hd$. The residuals

$$R_{01} = f(x) - f(x_+) - h\langle g_+, d\rangle - \tfrac{1}{2}\|g - g_+\|^2, \tag{23}$$

$$R_{10} = f(x_+) - f(x) + h\langle g, d\rangle - \tfrac{1}{2}\|g - g_+\|^2, \tag{24}$$

$$I = D\|g\|^2 - \|e\|^2 \tag{25}$$

are nonnegative. The first two are smooth convex interpolation inequalities; the third is the error condition. Also

$$f(x) - f_* \ge \tfrac{1}{2}\|\nabla f(x)\|^2. \tag{26}$$

For completeness, the interpolation inequality follows by subtracting the supporting affine function at one point and applying the smooth descent inequality to the resulting nonnegative convex function at the other point.

For the intermediate parameters (21), set

$$\lambda = \frac{q - D}{q(1-q)}, \qquad \beta = \frac{h}{2q}. \tag{27}$$

Direct expansion gives

$$\begin{aligned} f(x) - f(x_+) - \rho\|g_+\|^2 = (\lambda+1)R_{01} + \lambda R_{10} + \beta I \\ + \beta\|e + q((\lambda+1)g_+ - \lambda g)\|^2. \end{aligned} \tag{28}$$

The coefficient calculation uses precisely

$$D = q[1 - \lambda(1-q)], \quad h(\lambda+1)(1+q\lambda) = 2\lambda + 1, \quad \rho = \tfrac{h}{2}(\lambda+1)(1-q). \tag{29}$$

Once the parameter signs are checked below, (28) and (26) prove the intermediate upper bound.

In the left regime take $\lambda = 1$, $q = \delta$ and $\beta = h/(2\delta)$. The exact identity is

$$\begin{aligned} f(x) - f(x_+) - h(1-\delta)\|g_+\|^2 = 2R_{01} + R_{10} + \frac{h}{2\delta} I \\ + \frac{h}{2\delta}\|e + \delta(2g_+ - g)\|^2 \\ + \left(\frac{3}{2} - h(1+\delta)\right)\|g - g_+\|^2. \end{aligned} \tag{30}$$

Every coefficient is nonnegative for $0 \le h \le h_L$, including $h = 0$.

For the right regime let $r = h(1+\delta) - 1$. Then

$$r \in [q_R/\delta, 1], \quad \lambda = \frac{1-r}{r}, \quad \beta = \frac{h}{2\delta r}, \quad \rho = \frac{r^{-2} - 1}{2}.$$

One has

$$\begin{aligned} f(x) - f(x_+) - \rho\|g_+\|^2 = (\lambda+1)R_{01} + \lambda R_{10} + \beta I \\ + \beta\|e + \delta r((\lambda+1)g_+ - \lambda g)\|^2 + \xi\|g_+ + rg\|^2, \\ \xi = \frac{-\delta r^2 + (2+\delta)r - (1+\delta)}{2r^2(1+\delta)}. \end{aligned} \tag{31}$$

The numerator of ξ vanishes at $r = q_R/\delta$ and has positive derivative $2 + \delta - 2\delta r$ on the stated interval. Thus this certificate is nonnegative, including the upper endpoint. Adding (26) proves the remaining upper bound in (22).

3.2 Parameter signs and the global extremizer

Consider $T(q) = 2q - q^2 - D - \delta(1-q)$. It is strictly increasing on $[D, \delta]$, is negative at D, and is positive at δ. Its zero is q_R, so

$$D < q_R < \delta < 1, \qquad V \ge \delta(1-q) > 0, \qquad U = V + q - D > 0. \tag{32}$$

Also $T(\delta/2) = -3D/4 < 0$, hence $q_R > \delta/2$. Differentiation gives

$$h'(q) = -\frac{V^2 - (1-D)(D-q^2)}{(1-D)V^2} < 0, \tag{33}$$

because the numerator is at least $D(1-q)^2 - (1-D)(D-q^2) = (q-D)^2 > 0$. The endpoints are $h(\delta) = h_L$ and $h(q_R) = (1+q_R/\delta)/(1+\delta) = h_R$. This proves the bijection and $0 < h_L < h_R < 2/(1+\delta)$. In particular, the coefficients in (28) are positive.

Define

$$A = \frac{D+q^2-3q+1}{1-D}, \qquad B = \frac{D-q}{V}, \qquad C = \frac{(q-D)(2-q)}{1-D}, \tag{34}$$

and

$$a = \sqrt{C^2 - DA^2}, \qquad z = \sqrt{D-q^2}. \tag{35}$$

Both square roots are real, since

$$C^2 - DA^2 = \frac{V^2 - D(1-q)^2}{1-D} \geq 0. \tag{36}$$

Here $a = 0$ only at $q = q_R$, and $z = 0$ only at $q = \delta$. They therefore never vanish simultaneously.

For $a \geq 0$, let

$$\phi_a(u) = \begin{cases} u^2/2, & |u| \leq a, \\ a|u| - a^2/2, & |u| \geq a, \end{cases} \qquad \phi_0 \equiv 0. \tag{37}$$

The function

$$f(u,v) = \phi_a(u) + \tfrac{1}{2}v^2 \tag{38}$$

is globally convex and continuously differentiable. Its gradient is $(\mathrm{clip}_{[-a,a]}(u), v)$, hence is 1-Lipschitz. Its minimum is zero and is attained at the origin.

Take

$$x_0 = ((1+h(1-q))a, Az), \qquad d_0 = (1-q)(a,z). \tag{39}$$

The identities

$$A - B = h(1-q), \quad C = (1-q) - A, \quad -\lambda A + (\lambda+1)B = -C/q \tag{40}$$

give

$$x_1 = (a, Bz), \quad g_0 = (a, Az), \quad g_1 = (a, Bz), \quad e_0 = (-qa, Cz).$$

The first coordinate of x_0 is at least a. Moreover

$$\|e_0\|^2 - D\|g_0\|^2 = (q^2 - D)a^2 + (C^2 - DA^2)z^2 = 0.$$

Both interpolation residuals vanish, because the first coordinate travels in an affine part of ϕ_a and the second has unit quadratic curvature. The square in (28) vanishes by (40),

and $f(x_1) = \|g_1\|^2/2$. Thus the bound is attained. The construction remains valid if $A = 0$, since no division by A is used. In the open intermediate interval, $a, z > 0$ and $A - B > 0$, so g_0, g_1 are linearly independent.

For the left branch, use ϕ_1, $x_0 = 1 + h(1-\delta)$, and $d_0 = 1 - \delta$. For the right branch, use $f(x) = x^2/2$, $x_0 = 1$, and $d_0 = 1 + \delta$. These are the classical scalar examples in [1]. Their ratios attain the two outer expressions, completing the proof of Theorem 2.

3.3 The published root and the optimal step

The parameter λ in (27) is the largest real root of

$$\begin{aligned} 0 = {} & [(D-1)h^2 + 2h]\lambda^3 + [2(D-1)h^2 + 5h - 4]\lambda^2 \\ & + [(D-1)h^2 + 4h - 4]\lambda + h - 1. \end{aligned} \tag{41}$$

To check the root choice, note that $\lambda(q)$ increases strictly on $(D, 1)$:

$$\lambda'(q) = \frac{(q-D)^2 + D(1-D)}{q^2(1-q)^2} > 0.$$

If a larger real root $\lambda' > \lambda > 0$ existed, set $q' = [2\lambda' + 1 - h(\lambda' + 1)]/[h\lambda'(\lambda' + 1)]$. Its numerator is positive, because $h < 2$ and it is already positive at λ. Equation (41) implies $D = q'[1 - \lambda'(1 - q')]$, so q' is the unique positive root of that quadratic and lies in $(D, 1)$. Hence $q' > q$. But $h(q)$ is strictly decreasing throughout $[q_R, 1)$: above δ, the numerator in (33) is $V^2 + (1-D)(q^2 - D) > 0$. Thus $h(q') < h(q)$, contradicting the defining relation. This proves the root assertion.

Finally, (29) gives

$$\frac{1}{\rho + \frac{1}{2}} = \frac{2\lambda}{h\lambda^2 + (2h-1)\lambda + h - 1},$$

which is exactly the intermediate expression in [1, Theorem 2.3].

Corollary 3.1. *There is a unique $q_* \in (q_R, \delta)$ such that $\delta^2 = q_*^2(2 - q_*)$. The unique minimizer of $\mathcal{W}_1(h, \delta)$ on $[0, 2/(1+\delta)]$ is*

$$h_* = \frac{3 - 2q_*}{(2 - q_*)(1 + q_* - q_*^2)}, \qquad \mathcal{W}_1(h_*, \delta) = \frac{2(1 + q_* - q_*^2)}{(2 - q_*)^2}. \tag{42}$$

For the attaining instance (38)–(39) at this parameter,

$$\langle e_0, d_0 \rangle = 0, \qquad \langle e_0, g_0 \rangle = -\|e_0\|^2 = -\delta^2 \|g_0\|^2 < 0. \tag{43}$$

Proof. The left expression in (22) decreases with h and the right expression increases. In the intermediate interval,

$$\rho'(q) = \frac{D - q^2(2-q)}{q^2(1-D)}.$$

The function $q^2(2-q)$ increases strictly on $(0,1)$. Its root q_* for the displayed value D is below δ. It lies above q_R, since

$$q_*^2 - (2+\delta)q_* + D + \delta = (1-q_*)[\delta - q_*(2-q_*)] < 0.$$

Thus ρ has its unique maximum at q_*. Substitution proves (42). For the extremizer, direct simplification yields

$$\langle e_0, d_0 \rangle = \frac{(1-q)U}{1-D}[D - q^2(2-q)],$$

which vanishes at q_*. The error constraint is attained, and $g_0 \neq 0$, so (43) follows from $g_0 = d_0 - e_0$. ∎

The perpendicularity is to the computed direction. This distinction matters when interpreting the geometric observation in [1, Section 2.4]; it is not perpendicularity to the true gradient. We do not classify every possible extremizing function.

4 The exact asymptotic minimum-gradient curve

We now return to the minimum over all N updated points in (2). Define

$$h_{\max} = \frac{2}{1+\delta}, \qquad h_t = \frac{2}{1+\sqrt{\delta(2-\delta)}}. \tag{44}$$

Since $\delta < \sqrt{\delta(2-\delta)} < 1$, one has $1 < h_t < h_{\max}$. For $0 < h < h_{\max}$, set

$$\Gamma(h,\delta) = \begin{cases} h(1-\delta), & 0 < h \le h_t, \\ \dfrac{h(c^2-\delta^2)}{2u}, \quad c = \dfrac{2}{h} - 1, \quad u = 1 - \sqrt{1-c^2}, & h_t \le h < h_{\max}. \end{cases} \tag{45}$$

The second branch only uses $\delta < c \le \sqrt{\delta(2-\delta)} < 1$. Its parameters satisfy $c^2 = u(2-u)$ and $0 < u \le \delta$. At $h = h_t$, $u = \delta$, so both formulas agree. The function Γ is positive and continuous on the stable interval, and extends continuously by zero at its two endpoints.

Theorem 3. *For every fixed $0 < \delta < 1$ and $0 < h < h_{\max}$,*

$$\lim_{N\to\infty} N\mathcal{W}_N(h,\delta) = \frac{1}{\Gamma(h,\delta)}. \tag{46}$$

For every $N \ge 1$, the lower bound

$$\mathcal{W}_N(h,\delta) \ge \frac{1}{N\Gamma(h,\delta) + \frac{1}{2}} \tag{47}$$

is realized by an explicit scalar or two-dimensional example. The upper limit in (46) *is uniform over all admissible instances at each N.*

The exactness is asymptotic. In particular, (47) is not asserted to be the full finite-horizon curve. We first construct the lower examples and then give a finite-horizon upper estimate with the same leading coefficient.

4.1 Affine travel and transverse oscillation

In the first branch, use $f = \phi_1$, $x_0 = 1 + Nh(1-\delta)$, and $d_k = (1-\delta)g_k$. The actual trajectory is $x_k = 1 + (N-k)h(1-\delta)$ for $0 \le k \le N$. Every gradient is one, and $f(x_0) = Nh(1-\delta) + 1/2$. This gives (47).

For the second branch define

$$t = \frac{\delta^2 - u^2}{c^2 - \delta^2} \ge 0, \qquad b = 1 - u > 0, \tag{48}$$

and take the globally 1-smooth convex function $f(v, w) = \phi_1(v) + w^2/2$. Starting at $x_0 = (1 + Nhb, \sqrt{t})$, choose

$$d_k = \left(b, \frac{2}{h}(-1)^k \sqrt{t} \right). \tag{49}$$

Exact substitution in the recurrence gives

$$x_k = (1 + (N-k)hb, (-1)^k \sqrt{t}), \qquad 0 \le k \le N. \tag{50}$$

The first coordinate remains at least one. Consequently

$$g_k = (1, (-1)^k \sqrt{t}), \qquad e_k = (-u, c(-1)^k \sqrt{t}),$$

and

$$\|e_k\|^2 = u^2 + c^2 t = \delta^2(1+t) = \delta^2 \|g_k\|^2.$$

Every true gradient has the same norm. Moreover

$$f(x_0) = Nhb + \frac{1+t}{2}, \qquad \frac{hb}{1+t} = \frac{h(c^2 - \delta^2)}{2u} = \Gamma.$$

This proves (47) in the second branch as well. At the interface $t = 0$, so the construction becomes the scalar one. These examples also give terminal-gradient lower bounds; the upper bound in Theorem 3 concerns only the minimum gradient.

4.2 A local certificate with a telescoping correction

The residuals (23)–(25) allow a more flexible local identity than the uncorrected one-step bound. For suitable parameters, we have

$$\begin{aligned} f(x) - f(x_+) - \Gamma\|g_+\|^2 - \eta(\|g\|^2 - \|g_+\|^2) \\ = (\lambda+1)R_{01} + \lambda R_{10} + \beta I \\ + \beta\|e + \sigma((\lambda+1)g_+ - \lambda g)\|^2 + \zeta\|g - g_+\|^2. \end{aligned} \tag{51}$$

The right-hand multipliers are nonnegative. The coefficient η may have either sign; its two boundary terms will be retained.

For $0 < h \le h_t$, take

$$\sigma = \delta, \qquad \lambda = \frac{2 - h(1+\delta)}{2h\delta}, \qquad \beta = \frac{h}{2\delta}, \tag{52}$$
$$\eta = \frac{h}{2}(\delta - 1)(\lambda - 1), \qquad \zeta = \frac{h^2(1-\delta)^2 - 4h + 4}{8\delta h}, \qquad \Gamma = h(1-\delta).$$

Here $\lambda > 0$. The smaller positive root of the numerator of ζ is h_t, hence $\zeta \ge 0$. Expanding (51) verifies the identity.

For $h_t < h < h_{\max}$, use c, u from (45) and put

$$p = \frac{c}{1 + \sqrt{1 - c^2}} \in (0, 1). \tag{53}$$

Then

$$c = \frac{2p}{1+p^2}, \quad u = \frac{2p^2}{1+p^2}, \quad h = \frac{2(1+p^2)}{(1+p)^2}.$$

The parameters in (51) are

$$\sigma = u, \qquad \lambda = \frac{1-p}{2p}, \qquad \beta = \frac{h}{2u}, \qquad \zeta = 0, \tag{54}$$
$$\eta = \frac{p^4 + p^3 + 3p^2 - p - \delta^2(1+p^2)^2}{2p^2(1+p)^2},$$
$$\Gamma = \frac{4p^2 - \delta^2(1+p^2)^2}{2p^2(1+p)^2} = \frac{h(c^2 - \delta^2)}{2u}.$$

All right-hand multipliers are positive except $\zeta = 0$, and $\Gamma > 0$ since $c > \delta$. The rational substitutions above verify (51) directly.

4.3 Uniform control of both boundary terms

It remains to justify a sharp upper bound for the supremum in (2); a pointwise limit along one trajectory would not suffice. For the fixed stable h, δ, set

$$\alpha = \frac{h(1-\delta)^2}{1+\delta}\left(1 - \frac{h(1+\delta)}{2}\right) > 0. \tag{55}$$

The error condition gives

$$(1+\delta)\langle g, d\rangle - \|d\|^2 = \delta\|g\|^2 - (1-\delta)\langle g, e\rangle - \|e\|^2 \ge 0.$$

Smoothness and $\|d\| \ge (1-\delta)\|g\|$ therefore imply

$$f(x_k) - f(x_{k+1}) \ge \alpha\|g_k\|^2, \qquad \sum_{k=0}^{N-1}\|g_k\|^2 \le \frac{f(x_0)}{\alpha}. \tag{56}$$

Choose integers $m \ge 1$ and $N > 2m$. From (56), there exist

$$j \in \{1, \dots, m\}, \qquad \ell \in \{N - m, \dots, N - 1\}$$

such that $\|g_j\|^2, \|g_\ell\|^2 \le f(x_0)/(\alpha m)$. Sum (51) from j to $\ell-1$ and discard its nonnegative right side. Since f decreases and $f \ge 0$,

$$\begin{aligned}\Gamma \sum_{k=j+1}^{\ell} \|g_k\|^2 &\le f(x_j) - f(x_\ell) - \eta(\|g_j\|^2 - \|g_\ell\|^2) \\ &\le f(x_0)\left(1 + \frac{2|\eta|}{\alpha m}\right). \end{aligned} \tag{57}$$

The sum contains at least $N - 2m$ gradients from the index set in (2). Thus, uniformly over every admissible instance,

$$\mathcal{W}_N(h, \delta) \le \frac{1 + 2|\eta|/(\alpha m)}{\Gamma(h, \delta)(N - 2m)}. \tag{58}$$

Taking $m = \lfloor \sqrt{N} \rfloor$ for sufficiently large N proves the upper limit in (46). The lower examples prove the matching lower limit, completing Theorem 3.

5 Asymptotically optimal constant steps

The next result allows the selected constant step to depend on the horizon. Each individual run still uses one unchanged step.

Theorem 4. *Fix $0 < \delta < 1$. There is a unique $p_* \in (0, 1)$ satisfying*

$$\delta^2 = \frac{4p_*^3}{(1 + p_*^2)(1 + 2p_* - p_*^2)}. \tag{59}$$

Define

$$h_\infty = \frac{2(1 + p_*^2)}{(1 + p_*)^2}, \qquad C_\delta = \frac{1 + 2p_* - p_*^2}{2(1 - p_*)}. \tag{60}$$

Then $h_t < h_\infty < h_{\max}$, and

$$\lim_{N \to \infty} N \inf_{h \ge 0} \mathcal{W}_N(h, \delta) = C_\delta. \tag{61}$$

If $h_N \ge 0$ satisfies $N\mathcal{W}_N(h_N, \delta) \to C_\delta$, then $h_N \to h_\infty$. This includes exact minimizing steps whenever such minimizers exist.

Proof. Let $B(p) = (1 + p^2)(1 + 2p - p^2)$. The right side of (59) increases from zero to one on $(0, 1)$, because

$$\frac{d}{dp}\frac{4p^3}{B(p)} = \frac{4p^2(p + 1)^2(p^2 - 2p + 3)}{B(p)^2} > 0.$$

This proves existence and uniqueness. At this root, the substitutions (53) give

$$\frac{\delta^2}{c^2} = \frac{p(1 + p^2)}{1 + 2p - p^2} < 1, \qquad \frac{\delta^2}{u^2} = \frac{1 + p^2}{p(1 + 2p - p^2)} > 1.$$

The positive differences in the two comparisons are $(1-p)(1+p)^2$ and $(1-p)^2(1+p)$, respectively. Thus $u < \delta < c$, which places h_∞ in the open second branch.

In the p parametrization of that branch,

$$\frac{d\Gamma}{dp} = \frac{\delta^2 B(p) - 4p^3}{p^3(1+p)^3}. \tag{62}$$

The sign changes once, from positive to negative at p_*. Since h decreases strictly with p, Γ has its unique maximum there. Its first branch increases with h, so this maximum is global. Substitution yields

$$\Gamma_{\max} = \Gamma(h_\infty, \delta) = \frac{2(1-p_*)}{1+2p_*-p_*^2} = \frac{1}{C_\delta}. \tag{63}$$

For stable h, the actual lower instances in (47) give $\mathcal{W}_N(h,\delta) \geq [N\Gamma_{\max} + 1/2]^{-1}$. At $h = 0$, a scalar quadratic gives ratio two. For $h \geq h_{\max}$, the same quadratic with $d_k = (1+\delta)g_k$ has minimum-gradient ratio at least two, because the magnitude of its iteration factor is at least one. Hence the lower bound

$$\inf_{h\geq 0} \mathcal{W}_N(h,\delta) \geq \frac{1}{N\Gamma_{\max} + \frac{1}{2}}$$

holds on the whole noncompact step range. The fixed choice $h = h_\infty$ in Theorem 3 proves the matching upper limit, establishing (61).

Finally, an asymptotically minimizing sequence must be stable and positive for all sufficiently large N, since otherwise $N\mathcal{W}_N \geq 2N$ along a subsequence. The lower bound (47) forces $\Gamma(h_N, \delta) \to \Gamma_{\max}$. Extend Γ by zero at both endpoints of $[0, h_{\max}]$. Its continuity, the compactness of that interval, and the unique maximum proved above imply $h_N \to h_\infty$. ∎

The two optimal steps solve different questions. The step h_* in Corollary 3.1 is exactly optimal for one update. The step h_∞ in Theorem 4 is optimal in the asymptotic minimum-gradient problem. No convergence of the finite-horizon terminal-gradient optimal steps is implied. The finite-horizon examples start from points depending on N; they do not assert a nonconvergent infinite run from one fixed initialization.

6 A dimension-independent last-gradient bound fails

Define the terminal-gradient performance

$$\mathcal{W}_N^{\text{last}}(h,\delta) = \sup \frac{\|\nabla f(x_N)\|^2}{L(f(x_0) - f_*)}, \tag{64}$$

with the same class of instances as in (2). The supremum includes all finite dimensions. By Theorem 3, the minimum-gradient criterion at every $0 < h \leq 1$ satisfies $N\mathcal{W}_N(h,\delta) \to [h(1-\delta)]^{-1}$. The corresponding last-gradient quantity behaves differently.

Theorem 5. *Fix $0 < \delta < 1$ and $0 < h \le 1$, and put $r = 1 + h^2\delta^2/4 > 1$. For every integer $m \ge 1$, let $N = 2^{m-1}$. There is a globally convex, continuously differentiable, 1-smooth coercive function on $\mathbb{R}^{m+1}$, with unique minimizer zero, and an admissible run of (1) with $L = 1$ and normalized step h, such that*

$$f(x_0) - f_* = hN + \tfrac{1}{2}, \qquad \|\nabla f(x_N)\|^2 = r^m. \tag{65}$$

Consequently

$$\mathcal{W}_N^{\mathrm{last}}(h,\delta) \ge \frac{r^m}{hN + \frac{1}{2}} = \frac{rN^{\log_2 r}}{hN + \frac{1}{2}}, \qquad N = 2^{m-1}, \tag{66}$$

and

$$\limsup_{N\to\infty} N\mathcal{W}_N^{\mathrm{last}}(h,\delta) = +\infty. \tag{67}$$

Thus no finite constant $C(h,\delta)$ can bound $\mathcal{W}_N^{\mathrm{last}}(h,\delta)$ by $C(h,\delta)/N$ uniformly over this class, even for coercive objectives with unique minimizers.

Proof. Use coordinates $0, \dots, m$ and define positive amplitudes

$$a_0 = 1, \qquad a_j = \frac{h\delta}{2} r^{(j-1)/2} \quad (1 \le j \le m), \qquad f_m(x) = \sum_{j=0}^{m} \phi_{a_j}(x_j), \tag{68}$$

where ϕ_a is (37). The gradient is coordinatewise clipping, so f_m is globally convex, C^1 and 1-smooth. All amplitudes are positive. Therefore its unique minimizer is zero, and

$$f_m(x) \ge \left(\min_j a_j\right) \|x\|_1 - \tfrac{1}{2} \sum_{j=0}^{m} a_j^2$$

proves coercivity.

Split the N updates into m consecutive blocks, with lengths

$$n_j = 2^{m-j-1} \quad (1 \le j < m), \qquad n_m = 1. \tag{69}$$

For $m = 1$ there is only the last formula. Put $t_0 = 0$ and $t_j = \sum_{i=1}^{j} n_i$. Then

$$t_m = N, \qquad N - t_j = n_j \quad (j < m). \tag{70}$$

Initialize $x_0 = (hN + 1, 0, \dots, 0)$. If E_j denotes the coordinate unit vector with index j, prescribe

$$e_k = \frac{2a_j}{h} E_j, \qquad d_k = \nabla f_m(x_k) + e_k, \qquad t_{j-1} \le k < t_j. \tag{71}$$

These errors are specified before the run.

Coordinate zero equals $h(N-k) + 1$ throughout the run, staying at least one. Coordinate j is zero before its block. Its first active update sends it to $-2a_j$. Each subsequent active update subtracts $(2-h)a_j$, so

$$x_{t_{j-1}+s,j} = -[(2-h)s + h]a_j, \qquad 1 \le s \le n_j. \tag{72}$$

After that block the coordinate receives no error and moves right by ha_j per update as long as it stays at most $-a_j$. For $j < m$, the resulting formula is

$$x_{k,j} = -[(2-h)n_j + h - h(k-t_j)]a_j, \qquad t_j \le k \le N. \tag{73}$$

Its largest value occurs at N. By (70),

$$x_{N,j} = -[2(1-h)n_j + h]a_j \le -a_j,$$

since $n_j \ge 1$ and $h \le 1$. Thus the entire post-block formula is consistent with the clipped gradient. These formulas prove the actual trajectory by induction, including every junction contact. The last coordinate equals $-2a_m$ at time N, and hence

$$\nabla f_m(x_N) = (1, -a_1, \ldots, -a_m).$$

At the start of block j, the preceding coordinates have gradients $1, -a_1, \ldots, -a_{j-1}$, while coordinate j and all later coordinates have zero gradient. Their squared norm is

$$1 + \sum_{i=1}^{j-1} a_i^2 = r^{j-1}. \tag{74}$$

The error in (71) has norm $2a_j/h = \delta r^{(j-1)/2}$, so it satisfies the relative-error condition with equality at block entry. After the first update in that block, the squared gradient norm is r^j, while the error norm is unchanged. The condition is then strict. This verifies every prescribed direction.

Only coordinate zero contributes to the initial value: $f_m(x_0) = \phi_1(hN+1) = hN+1/2$. The final squared gradient is $1 + \sum_{j=1}^{m} a_j^2 = r^m$. This proves (65) and (66). Since $r > 1$, multiplying (66) by N and letting $m \to \infty$ proves (67). ∎

Every step in this theorem is strictly below $2/(1+\delta)$. The dimension is $m + 1 = 2 + \log_2 N$, so Theorem 5 does not exclude a bound with a dimension-dependent constant. It does not assert a slow infinite trajectory of one fixed instance. At $h = 1$, setting all subsequent errors to zero brings the displayed endpoint to its minimizer in two more updates: coordinate zero and all previously completed coordinates become zero after one update, while the last coordinate moves from $-2a_m$ to $-a_m$ and then to zero. For every h in the theorem, the minimum squared gradient among $x_1, \ldots, x_N$ is exactly r, so the normalized minimum-gradient ratio is $r/(hN + 1/2)$.

Taking $m = d - 1$ in the same construction gives, for every $d \ge 2$ and $0 < h \le 1$,

$$M_d(h, \delta) \ge \frac{(1 + h^2\delta^2/4)^{d-1}}{h + 2^{1-d}}. \tag{75}$$

This is a direct consequence of the preceding family. Together with Theorem 1, it gives

$$\frac{(1 + h^2\delta^2/4)^{d-1}}{h + 2^{1-d}} \le M_d(h, \delta) \le \frac{[2(1 + h(1+\delta))]^{2\lceil (1+4/c)^d \rceil}}{\alpha_h}, \tag{76}$$

with c, α_h as in (5)–(6). For fixed h, δ in this range, the lower bound is exponential in d and the upper bound is generally double exponential. Thus finiteness in every fixed dimension and failure of dimension-uniform control are both settled here; the dimension cost remains far from determined.

Gradient amplification itself is an established antecedent: [7, Theorem 2.8 and the discussion on pp. 39–41] studies it and obtains a non-tight last-gradient estimate by summing geometric weights. The obstruction above uses complete actual trajectories whose horizons and dimensions grow together. Its lower exponent is not claimed to be the sharp replacement for the failed $1/N$ rate.

7 Planar last-gradient estimates

Theorem 1 already gives the terminal rate in the plane throughout the stable interval. At smaller steps, the planar order of gradient rays yields an alternative explicit amplification constant. Its proof quantifies this geometry directly and also gives a continuous-time estimate. The bound is uniform over all objectives, starts and admissible errors in the stated planar class.

Theorem 6. *Let $f : \mathbb{R}^2 \to \mathbb{R}$ satisfy the assumptions of Section 1. Fix $0 < \delta < 1$ and*

$$0 < h \leq \frac{1}{1+\delta}.$$

Set

$$K_\delta = \frac{2\sqrt{\delta}}{1-\delta}, \qquad B_\delta = e^{2\pi K_\delta}, \qquad \alpha_h = h(1-\delta)\left(1 - \frac{h(1-\delta)}{2}\right). \tag{77}$$

Every run of (1) satisfies

$$\|\nabla f(x_j)\| \leq B_\delta \|\nabla f(x_i)\| \qquad (0 \leq i \leq j). \tag{78}$$

For every $N \geq 1$ and $f(x_0) > f_*$,

$$\frac{\|\nabla f(x_N)\|^2}{L(f(x_0) - f_*)} \leq \frac{1}{\frac{1}{2} + \alpha_h N / B_\delta^2}. \tag{79}$$

No boundedness assumption on the sublevels or trajectory is required.

The planar constant is not claimed to be sharp. Theorem 5 excludes a dimension-independent bound at every step in Theorem 6, whereas Theorem 1 gives a finite constant in every fixed dimension. The planar proof provides an alternative explicit constant, independent of h throughout its stated range; it is not needed to establish finiteness there.

7.1 A bound from newly visited gradient rays

We first isolate the planar argument. A continuous nonzero planar curve on a compact interval has a continuous real-valued angle lift. For an absolutely continuous curve, an absolutely continuous lift is obtained by integrating $\det(g, g')/\|g\|^2$ and fixing the initial angle.

Lemma 3. *Let $g : [0, T] \to \mathbb{R}^2 \setminus \{0\}$ be absolutely continuous, and write*

$$g(t) = e^{u(t)}(\cos\theta(t), \sin\theta(t)).$$

Suppose a later visit to the same oriented ray cannot increase the norm:

$$\theta(t) - \theta(s) \in 2\pi\mathbb{Z}, \quad s < t \quad \Longrightarrow \quad u(t) \le u(s). \tag{80}$$

Let $U(t) = \max_{0\le s\le t} u(s)$. Assume that, almost everywhere on the set $\{u = U,\ u' > 0\}$,

$$u'(t) \le K|\theta'(t)| \tag{81}$$

for a fixed $K \ge 0$. Then

$$\max_{0\le t\le T} \|g(t)\| \le e^{2\pi K}\|g(0)\|.$$

Proof. Let $a(t) = \min_{s\le t}\theta(s)$, $b(t) = \max_{s\le t}\theta(s)$ and $w(t) = b(t) - a(t)$. The running extrema of an absolutely continuous scalar function are absolutely continuous: their variation on an interval is bounded by the integral of the absolute derivative there. Thus U, w are absolutely continuous and nondecreasing.

Work at a time where all these functions are differentiable. If $U' > 0$, then $u = U$ and $u' = U'$: a strict gap below the running maximum would make U locally constant, and the derivative of $U - u$ at its zero is zero. If the current ray had been visited at some $s < t$, (80) would force equality throughout

$$u(t) \le u(s) \le U(s) \le U(t) = u(t).$$

Then U would be constant on $[s, t]$, contradicting $U'(t) > 0$. Therefore a growing record must visit a new ray.

Continuity shows that an angle strictly between $a(t)$ and $b(t)$ was already visited. Hence growing records occur only at an extreme of the lifted angle. At $\theta = b$, differentiation of $b - \theta$ at its zero gives $\theta' = b' \ge 0$; at $\theta = a$, it gives $\theta' = a' \le 0$. In either case $|\theta'| \le w'$. Once $w \ge 2\pi$, every ray has already been visited: at an angular extreme use the earlier angle differing by 2π, and in the interior use continuity. Thus, almost everywhere,

$$0 \le U' \le Kw'\mathbf{1}_{\{w<2\pi\}}. \tag{82}$$

Equivalently, $U - K\min\{w, 2\pi\}$ is nonincreasing. Integration and $w(0) = 0$ give $U(T) - U(0) \le 2\pi K$, which proves the result. ∎

For gradients of one smooth convex function, the ray condition follows from objective descent. Indeed, if $f(y) \le f(x)$ and $\nabla f(y) = \lambda \nabla f(x) \neq 0$ with $\lambda > 1$, cocoercivity gives

$$\langle \nabla f(x), y - x \rangle \ge \frac{\lambda - 1}{L} \|\nabla f(x)\|^2 > 0.$$

Convexity would imply $f(y) > f(x)$, a contradiction. Consequently any path along which f is nonincreasing satisfies (80).

7.2 The actual polygonal path and the endpoint

Proof of Theorem 6. Normalize $L = 1$. Write $g_k = \nabla f(x_k)$ and $d_k = g_k + e_k$. On one actual step, put

$$G(q) = \nabla f(x_k - q d_k), \qquad 0 \le q \le h.$$

The error inequality and cocoercivity respectively give

$$\left\| g_k - \frac{d_k}{1 - \delta^2} \right\| \le \frac{\delta}{1 - \delta^2} \|d_k\|, \tag{83}$$

$$\left\| G(q) - g_k + \frac{q}{2} d_k \right\| \le \frac{q}{2} \|d_k\|. \tag{84}$$

For (84), expand $\|G(q) - g_k\|^2 \le -q \langle G(q) - g_k, d_k \rangle$. Adding the two balls places $G(q)$ in the ball with center $[1/(1 - \delta^2) - q/2] d_k$ and radius $[\delta/(1 - \delta^2) + q/2] \|d_k\|$. These balls are nested as q increases. At $q = 1/(1 + \delta)$ the center coefficient and radius coefficient both equal $1/[2(1 - \delta)]$. Hence, including at the step endpoint,

$$\left\| G(q) - \frac{d_k}{2(1 - \delta)} \right\| \le \frac{\|d_k\|}{2(1 - \delta)}, \qquad \langle G(q), d_k \rangle \ge (1 - \delta) \|G(q)\|^2. \tag{85}$$

The second inequality is just the squared first inequality. In particular, f is nonincreasing along the entire spatial segment.

Join all the actual steps by their affine interpolations and restrict this path to any finite interval of vertices $i, \dots, j$. If the final gradient is zero, (78) is immediate. Otherwise the gradient is nonzero everywhere on this path: a zero gradient would attain f_*, after which the objective could not decrease further and every later gradient would be zero. A zero-gradient vertex is stationary because its error is zero.

The gradient path is absolutely continuous. At almost every time, monotonicity of ∇f, applied to difference quotients, gives

$$\langle g'(t), x'(t) \rangle \ge 0.$$

On the interior of an edge this means $\langle g', d_k \rangle \le 0$; no Hessian is assumed to exist. Write $g = rn$, $n = (\cos\theta, \sin\theta)$, and let Jn be the counterclockwise rotation of n by $\pi/2$.

At a running norm record on this restricted path, $r \ge \|g_k\|$ because the current edge starts at an earlier vertex. Since $\|d_k\| \le (1 + \delta)\|g_k\|$, (85) gives

$$\frac{\langle g, d_k \rangle}{r \|d_k\|} \ge \frac{1 - \delta}{1 + \delta}. \tag{86}$$

Resolve $d_k = An + CJn$. Then $A > 0$ and $|C|/A \le 2\sqrt{\delta}/(1-\delta) = K_\delta$. The inequality $\langle g', d_k\rangle \le 0$ therefore yields, at these record times,

$$(\log r)' = \frac{r'}{r} \le -\frac{C}{A}\theta' \le K_\delta|\theta'|.$$

The objective is nonincreasing on the full path, so the ray condition holds. Lemma 3, restarted at vertex i, proves (78). The argument requires a uniform angle only at norm records; it remains valid when the all-points angle degenerates at $h = 1/(1+\delta)$.

It remains to sum descent. Since $h \le 1/(1+\delta) < 1$, smoothness gives

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\ge \left(h - \frac{h^2}{2}\right)\|g_k\|^2 + h(1-h)\langle g_k, e_k\rangle - \frac{h^2}{2}\|e_k\|^2 \\ &\ge \alpha_h\|g_k\|^2. \end{aligned} \tag{87}$$

Use $\|g_k\|^2 \ge B_\delta^{-2}\|g_N\|^2$ for $0 \le k < N$ and the terminal bound $f(x_N) - f_* \ge \|g_N\|^2/2$. Their sum is

$$f(x_0) - f_* \ge \left(\frac{1}{2} + \frac{\alpha_h N}{B_\delta^2}\right)\|g_N\|^2.$$

Restoring L proves (79). ∎

7.3 Continuous relative-error flow

The same planar mechanism has a direct continuous version with a smaller constant.

Proposition 7.1. Let $f : \mathbb{R}^2 \to \mathbb{R}$ satisfy the assumptions of Section 1, and let x be a locally absolutely continuous solution of

$$x'(t) = -\nabla f(x(t)) - e(t), \qquad \|e(t)\| \le \delta\|\nabla f(x(t))\| \quad \text{almost everywhere,}$$

with measurable e. Put $\kappa_\delta = \delta/\sqrt{1-\delta^2}$. Then, for all $0 \le s \le t$,

$$\|\nabla f(x(t))\| \le e^{2\pi\kappa_\delta}\|\nabla f(x(s))\|. \tag{88}$$

For every $T > 0$,

$$\|\nabla f(x(T))\|^2 \le \frac{e^{4\pi\kappa_\delta}}{(1-\delta)T}\big(f(x(0)) - f_*\big). \tag{89}$$

Proof. Write $g = \nabla f(x)$. On a compact interval where $g \ne 0$, the chain rule gives $f' = -\|g\|^2 - \langle g, e\rangle \le -(1-\delta)\|g\|^2$. Thus the ray condition holds. Monotonicity, differentiated only along the absolutely continuous paths, gives

$$\langle g', g\rangle \le -\langle g', e\rangle \le \delta\|g'\|\|g\|.$$

Writing $g = r(\cos\theta, \sin\theta)$, whenever $r' > 0$ this implies

$$r' \le \delta\sqrt{(r')^2 + r^2(\theta')^2}, \qquad (\log r)' \le \kappa_\delta|\theta'|.$$

Lemma 3 proves (88) on every such interval. If a zero gradient is reached at x_*, the estimate $\|x'\| \le (1+\delta)L\|x-x_*\|$ and Gronwall's inequality show that the solution remains there. This also covers zero initial gradients.

Finally, apply (88) from each $s \in [0,T]$ to T, and integrate the descent inequality:

$$f(x(0)) - f_* \ge (1-\delta)\int_0^T \|g(s)\|^2\,ds \ge (1-\delta)Te^{-4\pi\kappa_\delta}\|g(T)\|^2.$$

This proves (89). ∎

The discrete ray argument includes the endpoint $h = 1/(1+\delta)$, while Theorem 1 settles the planar unit step and the full admissible constant-step range by a different mechanism. The continuous statement retains its smaller explicit amplification factor. Sharp terminal constants and their dependence on dimension remain undetermined.

8 Conclusions and remaining constants

For arbitrary relative gradient errors below one, the terminal squared-gradient rate depends essentially on whether the dimension is fixed. Every finite dimension admits a function-gap-normalized $O(1/N)$ guarantee throughout the exact open stability interval. The last-exit argument supplies the missing uniform control of gradient amplification, including the planar unit step. The growing-dimensional Huber family shows that such control cannot be made independent of dimension at $0 < h \le 1$.

The exact asymptotic minimum-gradient coefficient and its optimal constant step address a separate performance question. The one-step curve is retained with attribution of the public intermediate-attainment result. The planar ray proof remains useful for its explicit smaller-step constant and continuous counterpart. The principal quantitative gaps are the sharp fixed-dimensional terminal constants, the growth of those constants with dimension, and the exact finite-horizon minimum-gradient curve. In particular, the exponential lower bound and generally double-exponential upper bound in (76) do not determine a sharp dimension dependence.

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